

Under the Hood of the Web

Request Structures, DNS Records, and File Mapping

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1. The Anatomy of a GET Request

When you type a URL and hit Enter, your browser sends a text message to the server.

The Structure:

- ❶ **Request Line:** What you want + Version.
- ❷ **Headers:** Info about you (Browser, Language).
- ❸ **Body:** (Empty for GET).

The Actual Text Sent:

```
GET /courses/it213.html HTTP/1.1
Host: www.concordia.ab.ca
User-Agent: Chrome/90.0
Accept-Language: en-US
```

Translation: "Hey Server, please read the file 'it213.html' inside the 'courses' folder and send it to me."

2. The Anatomy of a POST Request

When you fill out a form and click "Submit", the structure changes slightly.

The Difference:

- We need to **send data** to the server.
- We use the **Body** section (below the headers).

The Actual Text Sent:

```
POST /login.php HTTP/1.1
Host: www.concordia.ab.ca
Content-Type: application/x-www-form-urlencoded

username=student&password=secret123
```

Translation: "Hey Server, here is some private data (username/password). Please process it with 'login.php'."

3. The DNS "Table" Concept

How does the computer know where to send those requests? It looks at a ****DNS Table**** (Resource Records).

Imagine a simple spreadsheet inside the DNS Server:

Name (Key)	Value (Result)	Type	TTL
www.google.com	142.250.189.68	A	300s
mail.google.com	172.217.1.5	A	300s
shop.google.com	www.google.com	CNAME	600s

The Simple Query Process:

- 1 **Query:** "I need the **Type A** address for **www.google.com**."
- 2 **Lookup:** Server scans the "Name" column.
- 3 **Answer:** "The Value is **142.250.189.68**."

4. Mapping URL to Server Hard Drive

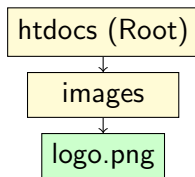
Once the request reaches the correct IP, how does the server find the file?

The URL: `http://www.mysite.com/images/logo.png`

The Server's Hard Drive:

The web server software
(Apache/Nginx) has a designated
"**Root Folder**":

`C:\xampp\htdocs\`



The Translation: The server simply combines the **Root** + **URL Path**:

`C:\xampp\htdocs` + `\images\logo.png`

5. Evolution: From FTP to HTTP

The Old Way: FTP (File Transfer Protocol)

In the past, if you wanted to move a file to a server, you needed a special client (like FileZilla).

- You had to log in.
- You manually dragged and dropped files.
- It was a separate "session" just for files.

The Modern Way: HTTP everywhere

Today, we rarely use manual FTP. HTTP does it all in the background!

- **Downloading?** That's just a GET request.
- **Uploading a photo?** That's just a POST request.

The Web Browser has become the universal file transfer tool.